HOCHSCHULE
KONSTANZ
UNIVERSITY OF
APPLIED SCIENCES

Software Architecture

Architecture Patterns

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HTWG Konstanz

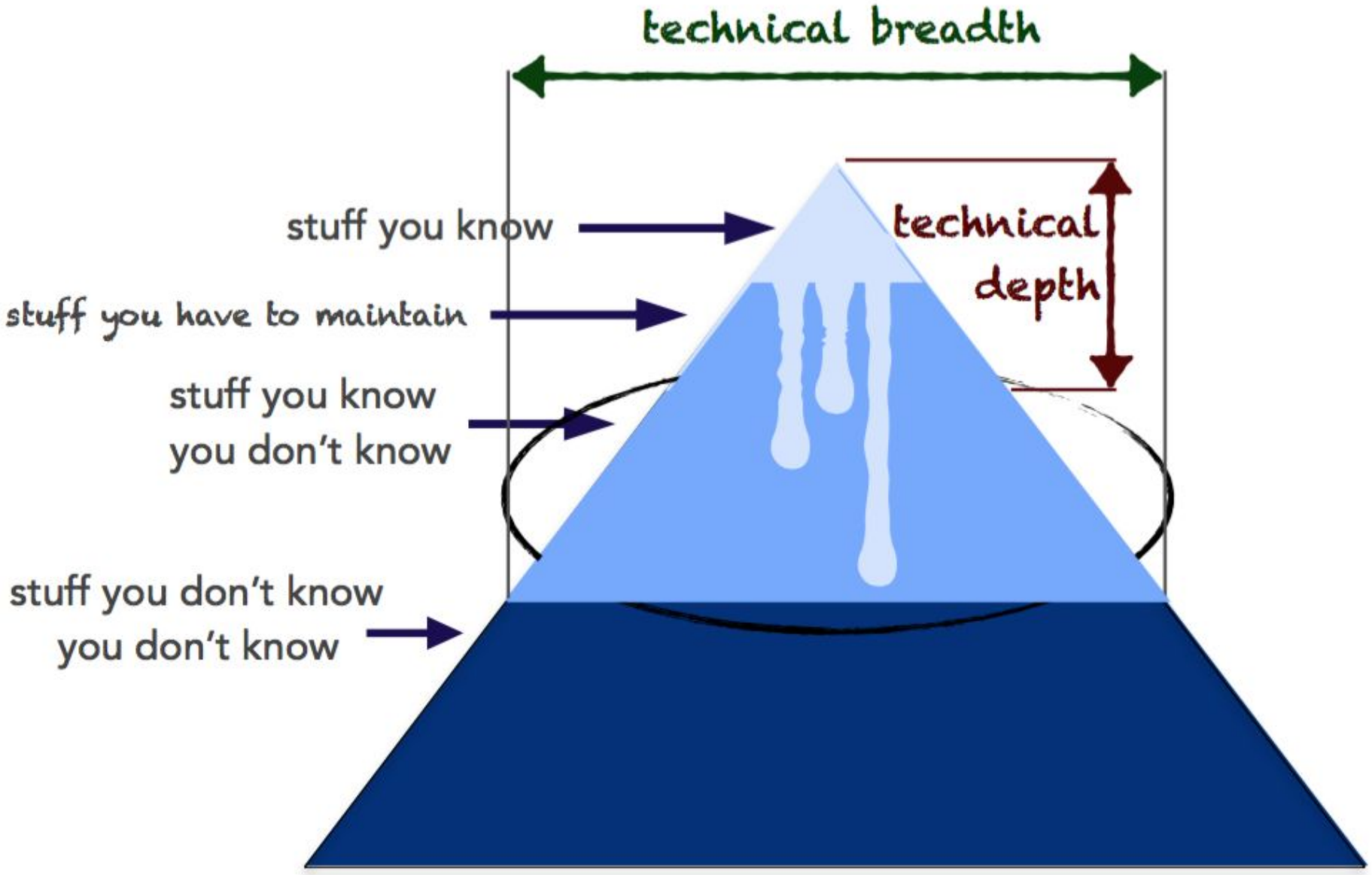
Definition of Architecture

- Architecture is the stuff that is hard to change
  - „Architecture is about the important stuff. Whatever that is.“

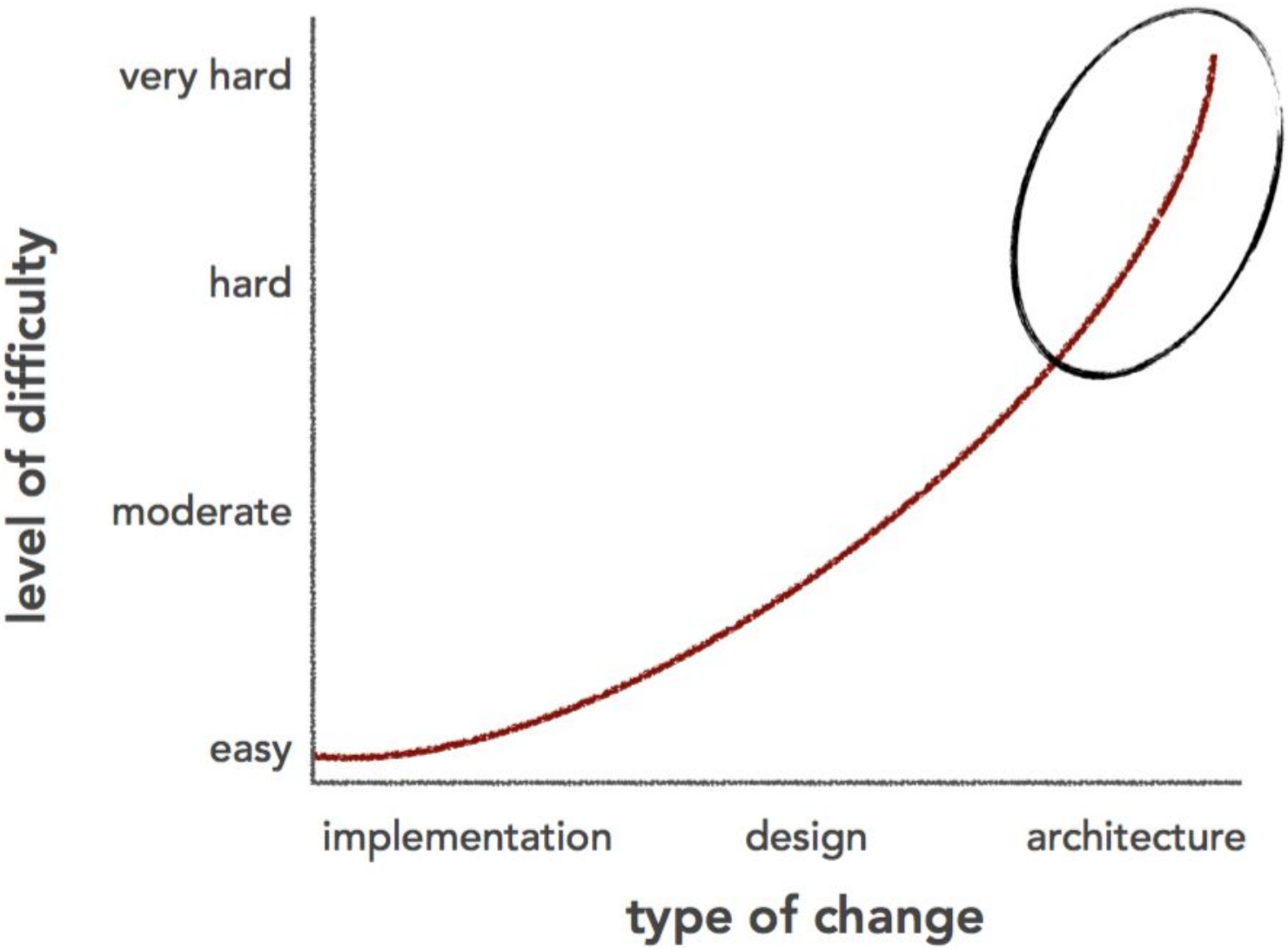
    Martin Fowler

- Architecture is the balance between benefits and tradeoffs

Knowlege Triangle



Architecture Decisions



Changes in the landscape of Software Architecture

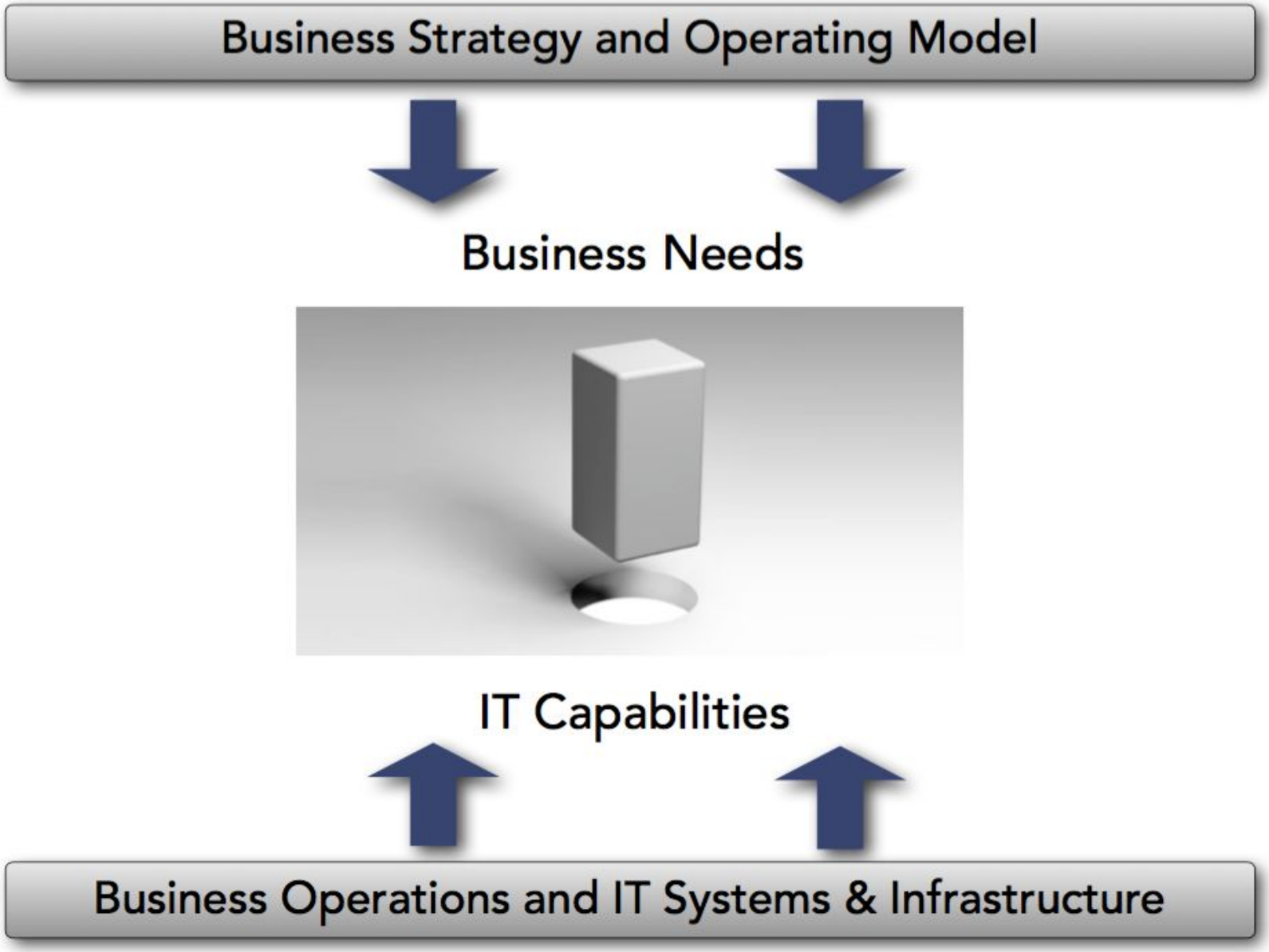
- Agile Development
- Continuous Delivery
- Continuous Deployment
- DevOps Revolution
  - Virtual Machines
  - Puppet/Chef
  - Docker
- Evolutionary Architecture

Levels of Architecture

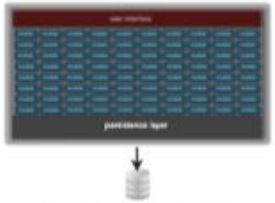
- System Context
- System
- Module
- Component
- Class

- Enterprise Architecture
- Integration Architecture
- Application Architecture
- Component Design
- Implementation Design

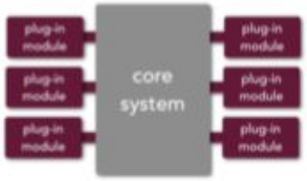
Enterprise Architecture



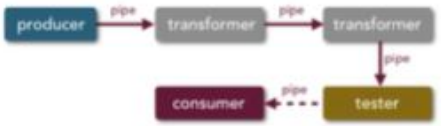
Architecture Patterns



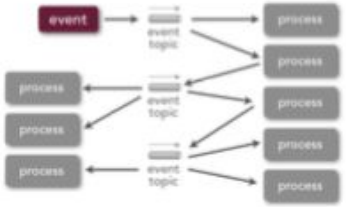
layered



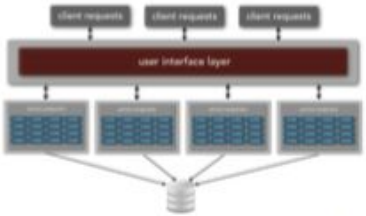
microkernel



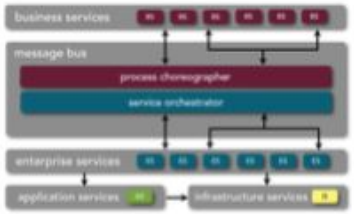
pipeline



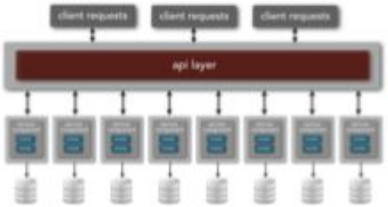
event-driven



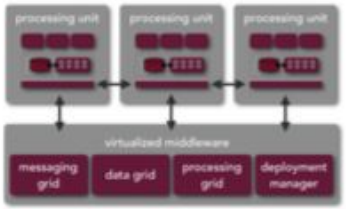
service-based



service-oriented

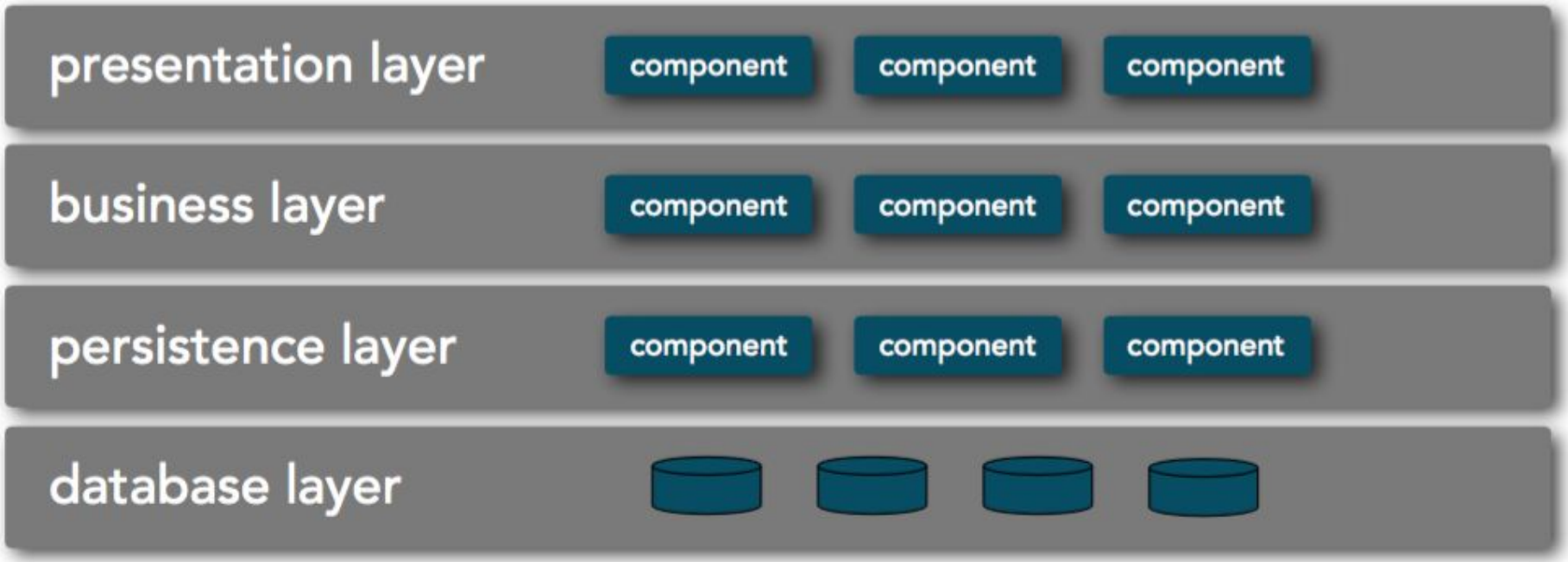


microservices

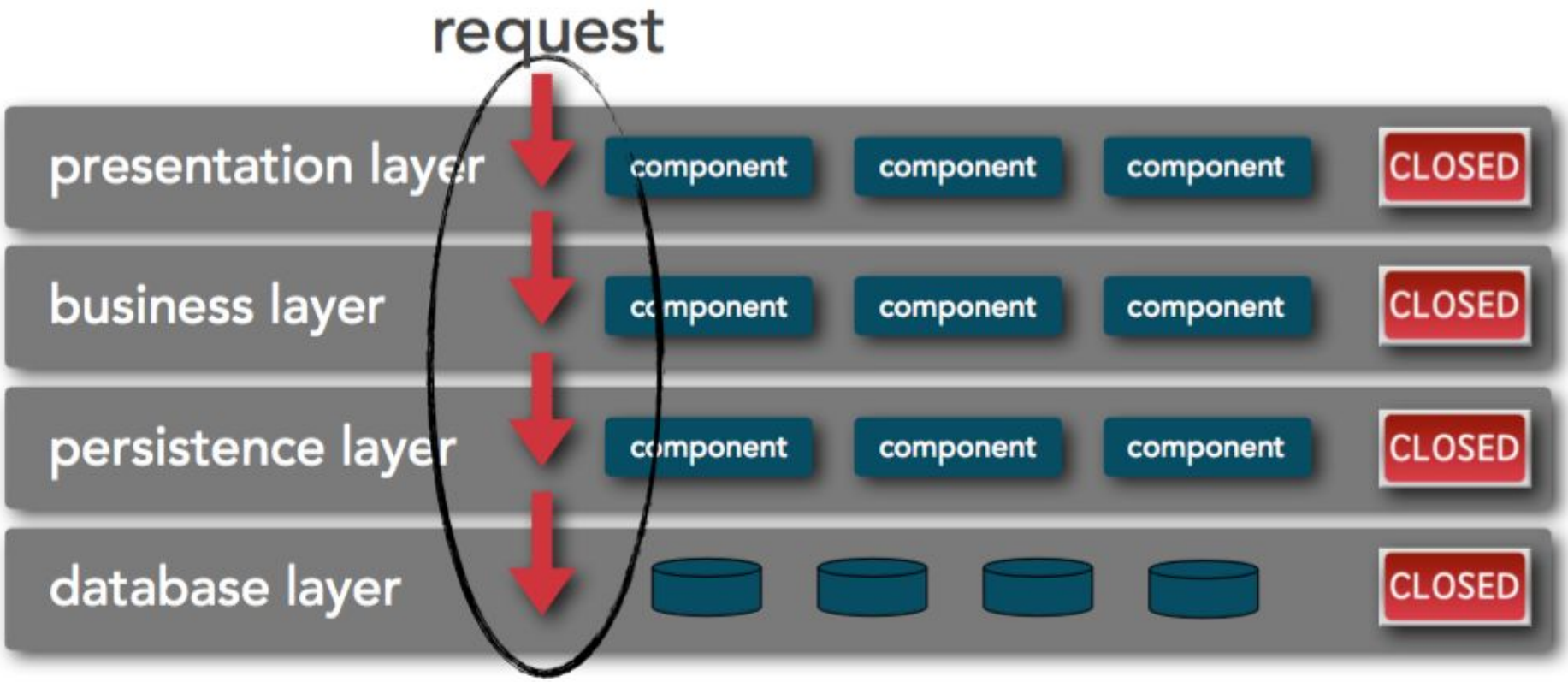


space-based

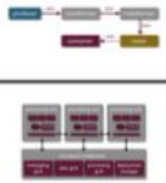
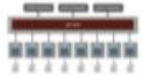
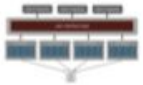
Layered Architecture



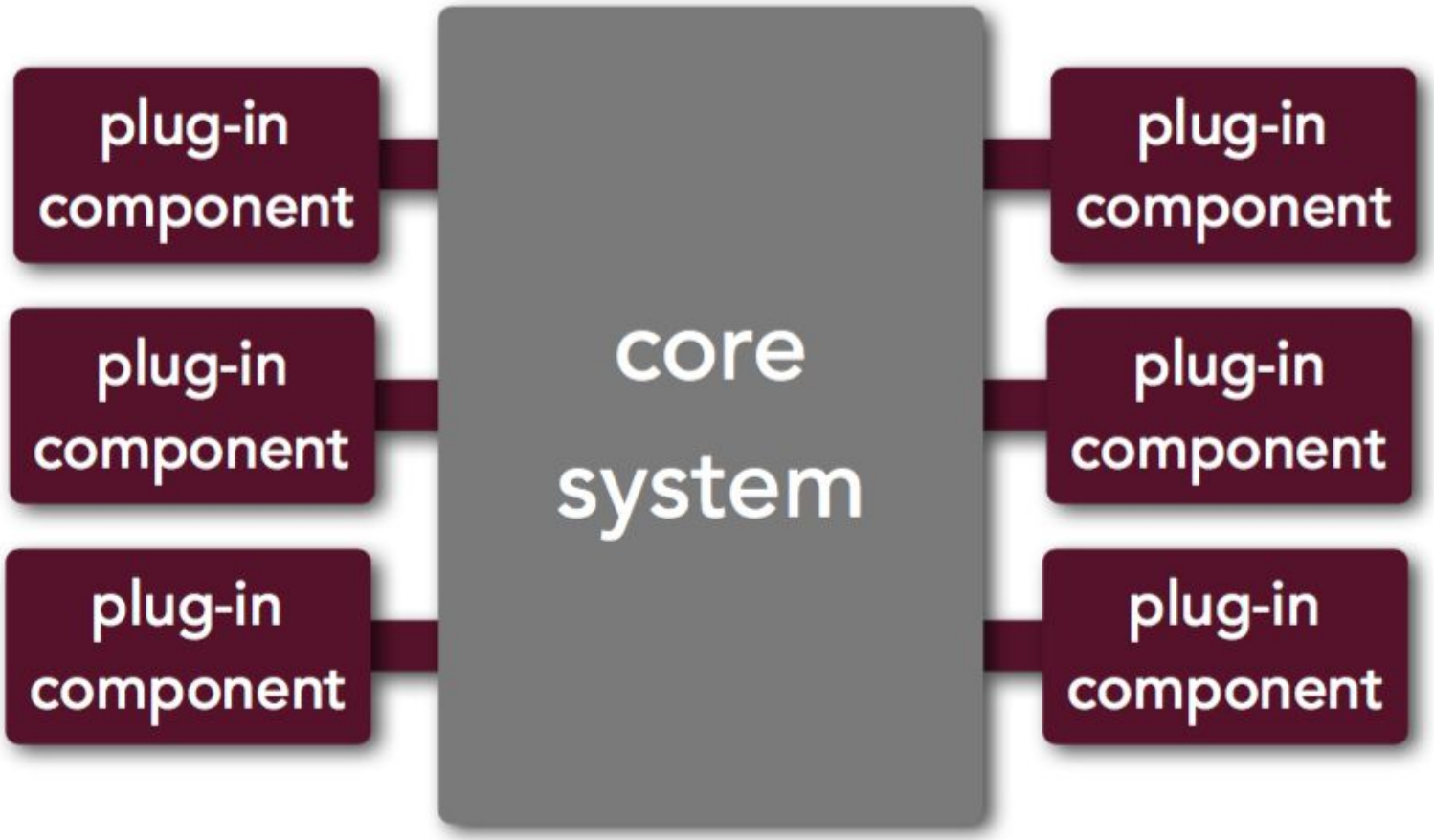
Closed Layer



Layered Architecture

	agility	deployment	testability	performance	scalability	simplicity	cost
	👎	👎	👍	👎	👎	👍	$
							
							
							

Microkernel or Plug-in Architecture



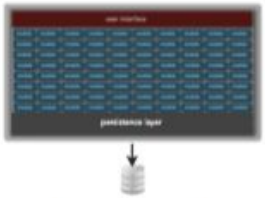
Eclipse as Microkernel

Microkernel Architecture

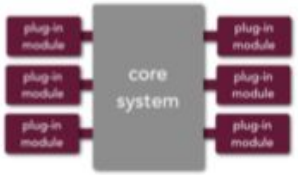
	agility	deployment	testability	performance	scalability	simplicity	cost
	👎	👎	👍	👎	👎	👍	$
	👍	👍	👍	👎	👎	👎	$$

Architecture Pattern Classification

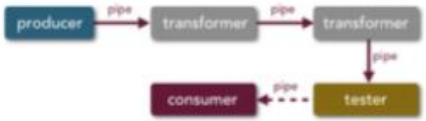
monolithic



layered

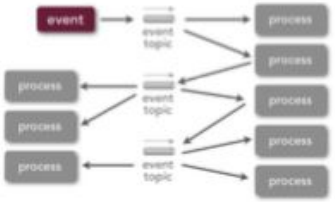


microkernel

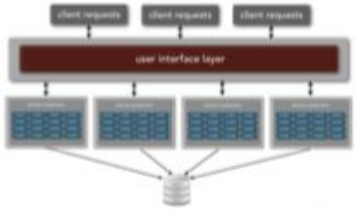


pipeline

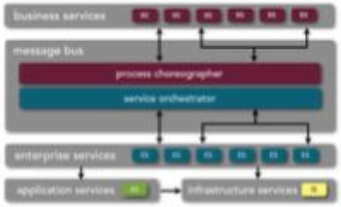
distributed



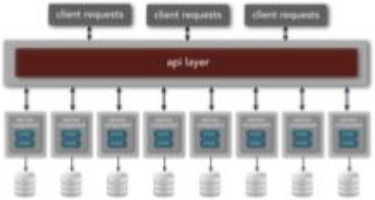
event-driven



service-based



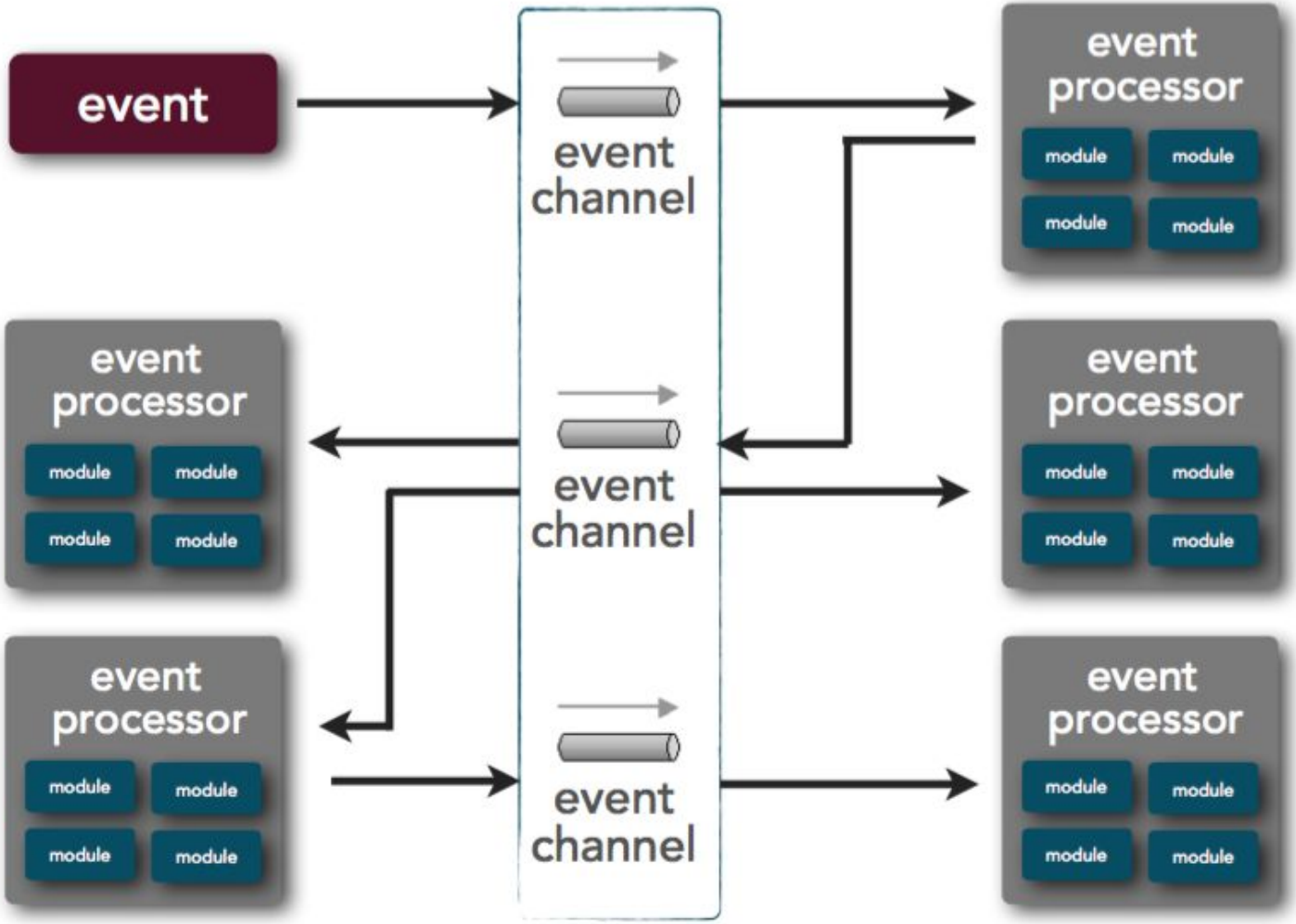
service-oriented



microservices

space-based

Event-driven Architecture

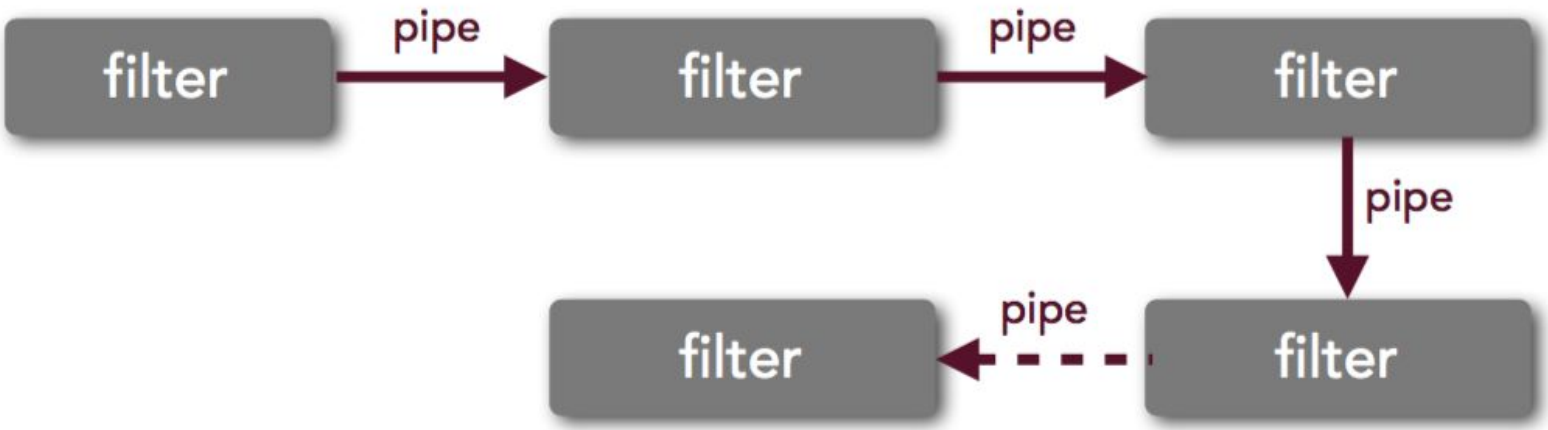


Event-driven Architecture

	agility	deployment	testability	performance	scalability	simplicity	cost
	👎	👎	👍	👎	👎	👍	$
	👍	👍	👍	👎	👎	👎	$$
	👍	👍	👎	👍	👍	👎	$$$

Pipeline Architecture

- Also Pipes and Filter Architecture

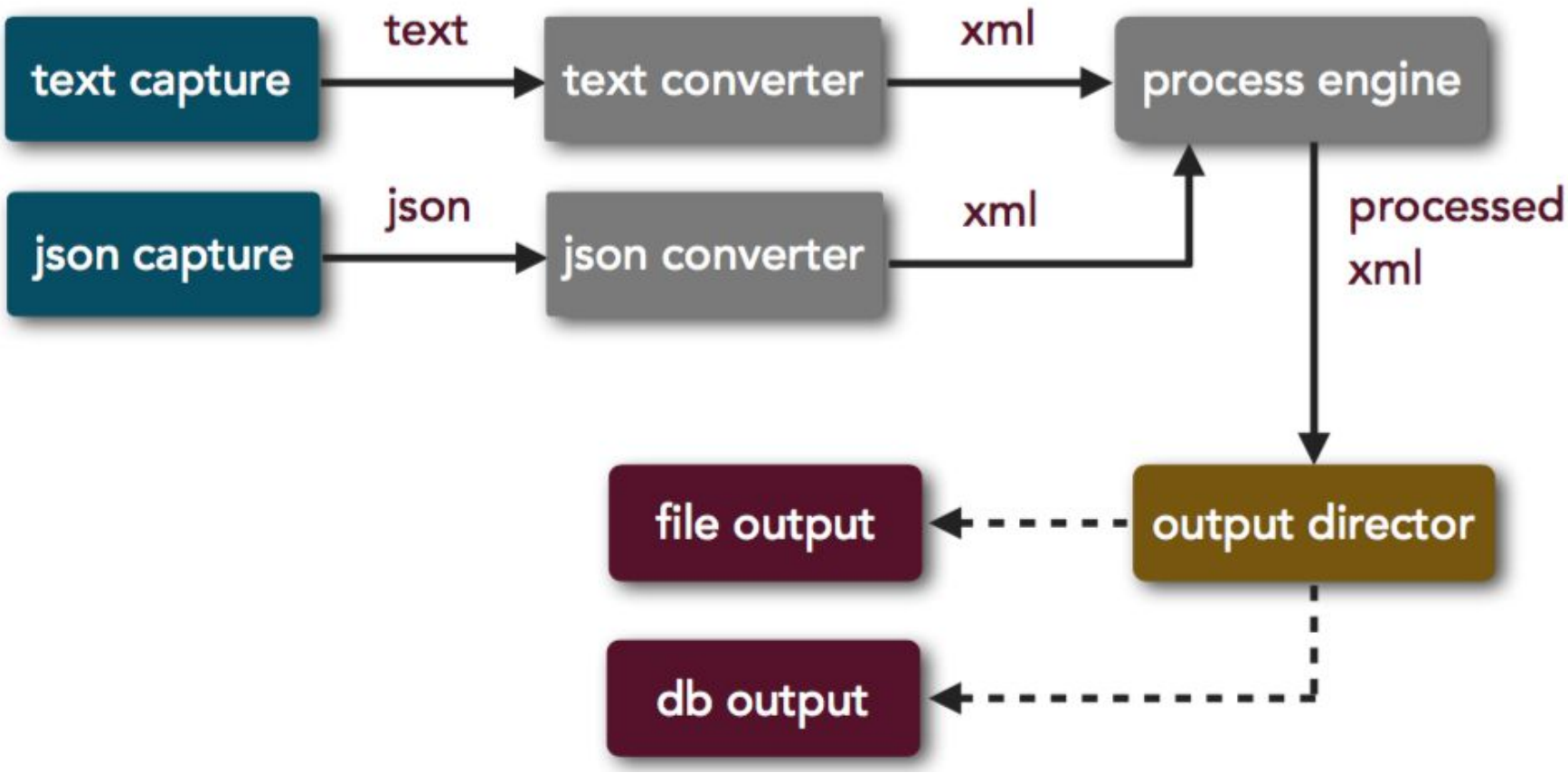


Pipes and Filters

- Pipes
  - Pipes are a transport mechanism for data. This can be the file system, a text format, a point-to-point communication or a message bus
- Filter
  - Filter are transformations, transforming an input data type into an output datatype.
  - The Payload can have many types, often it is text, xml, JSON, bytes or files

Example Pipeline

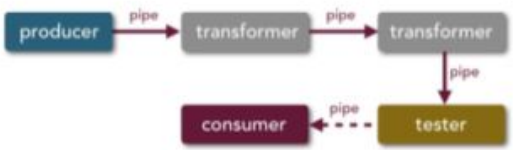
- Capture data in multiple formats, process the data and produce multiple outputs

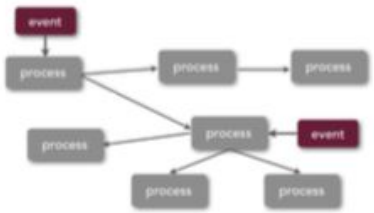


Pipeline Architecture

	agility	deployment	testability	performance	scalability	simplicity	cost
	👎	👎	👍	👎	👎	👍	$
	👍	👍	👍	👎	👎	👎	$$
	👍	👍	👎	👍	👍	👎	$$$
	👍	👎	👍	👎	👎	👍	$

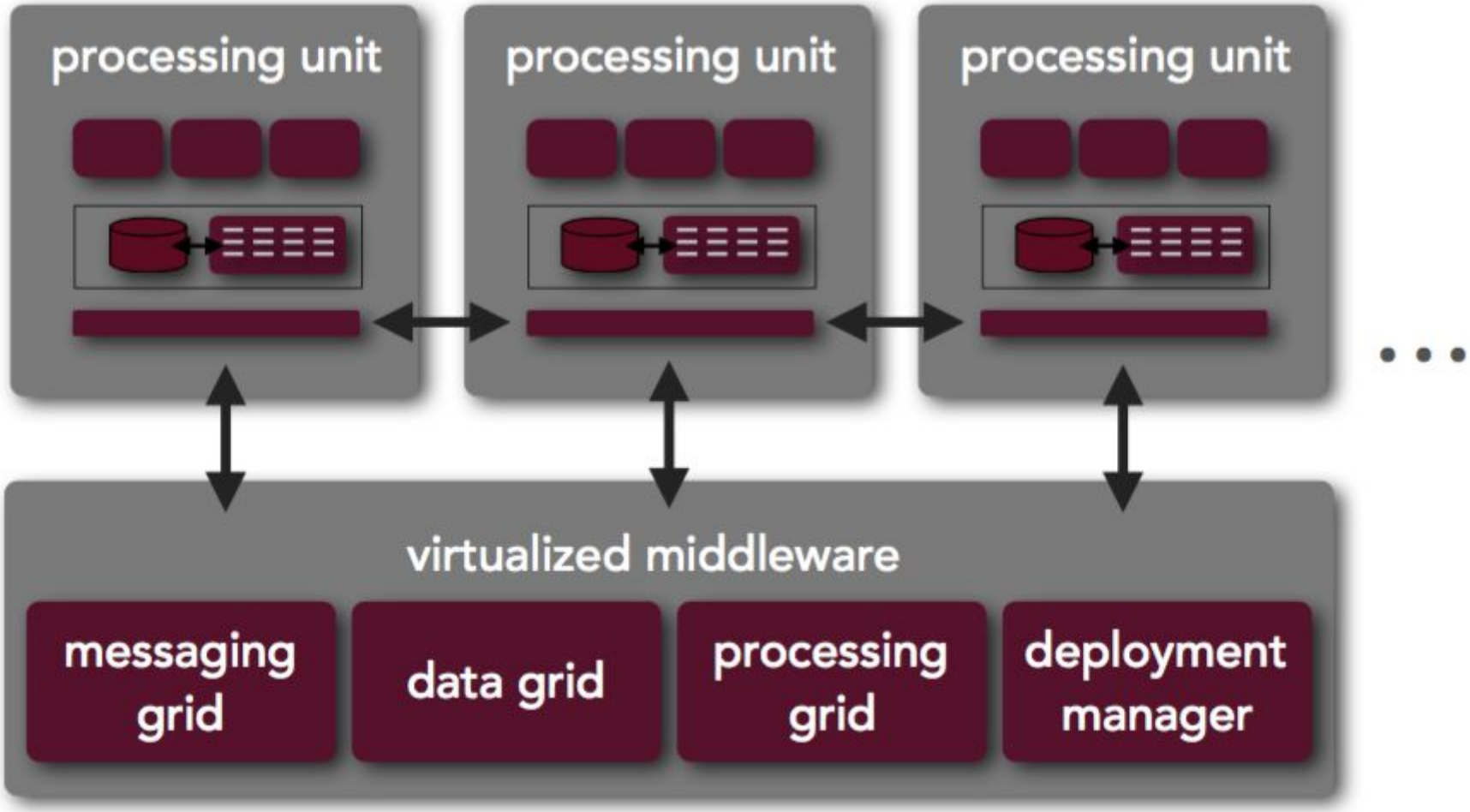
Pipeline vs Event-driven



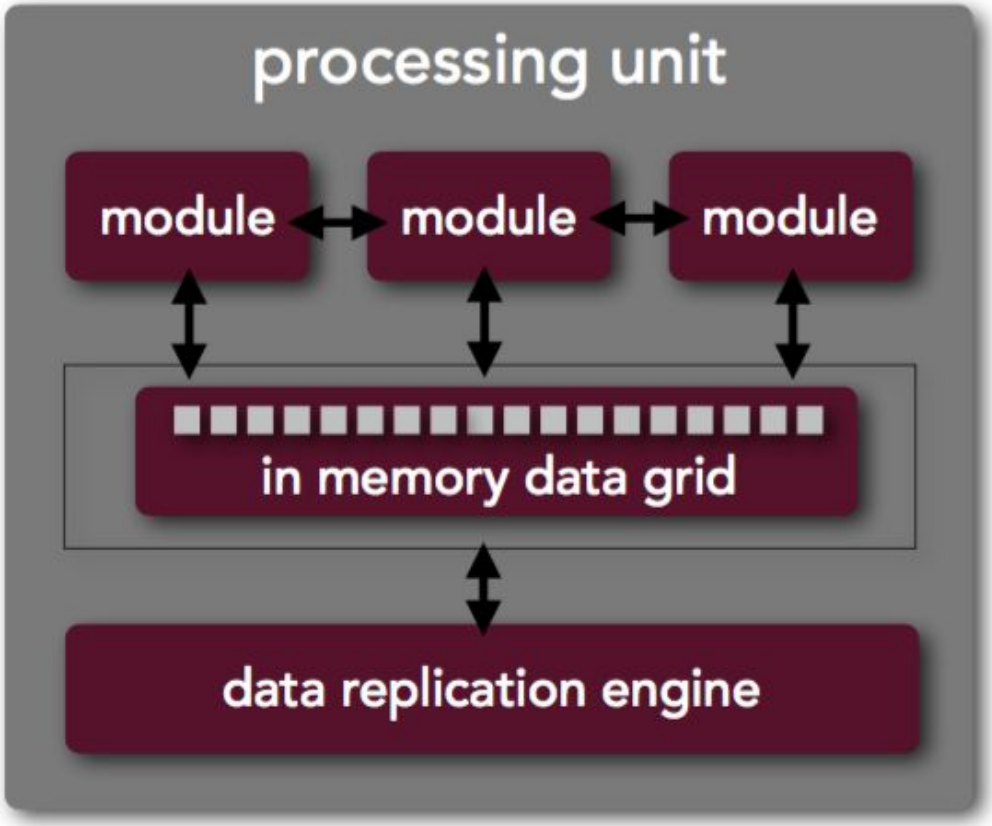


synchronous data filtering	asynchronous event processing
always unidirectional	can be request/reply
simple single purpose filters	complex multi-purpose processors

Space-based Architecture



Processing Unit



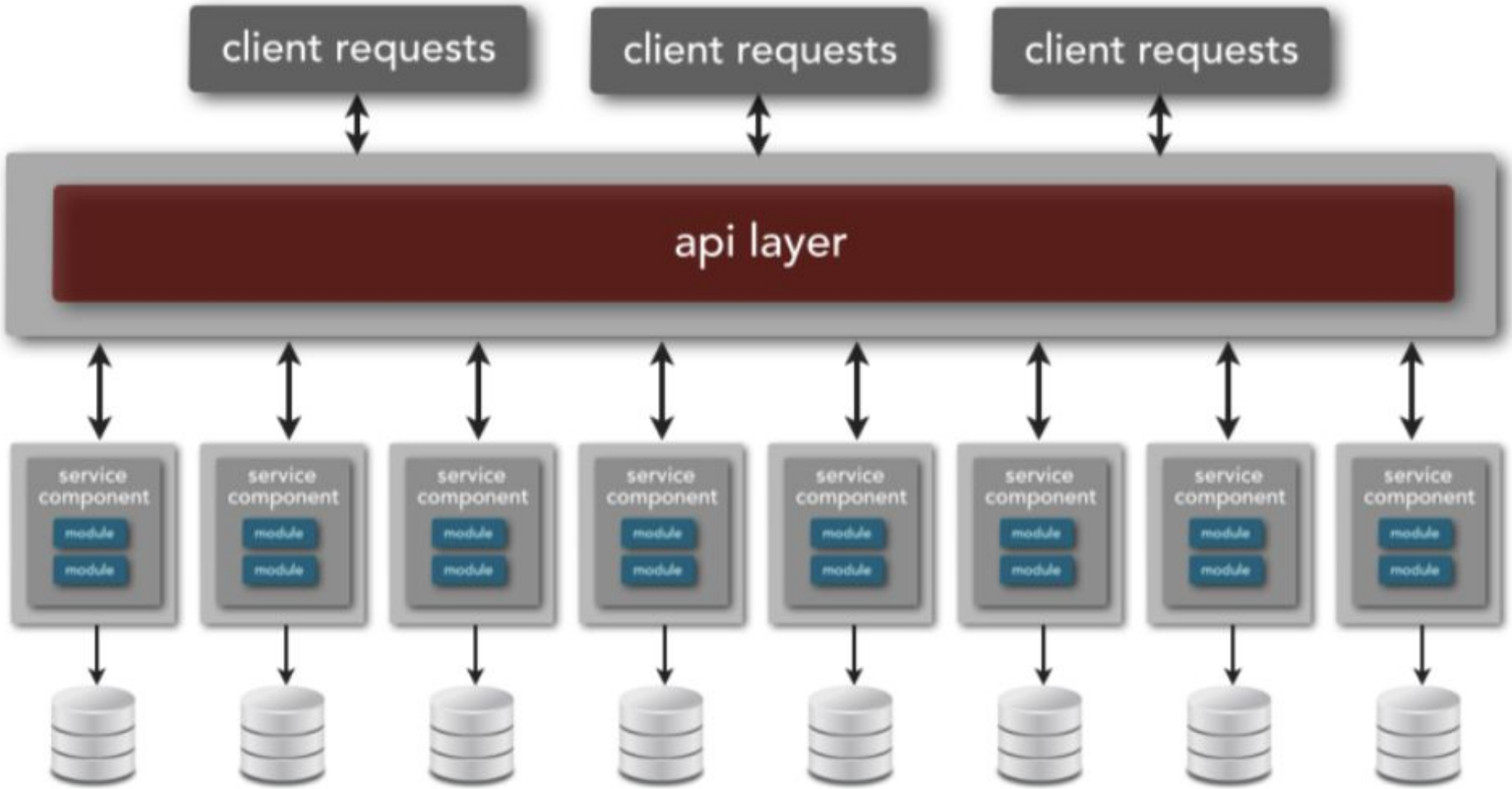
Space-based Architecture

- All about scalability
- Good for Applications with variable load
  - Peak times

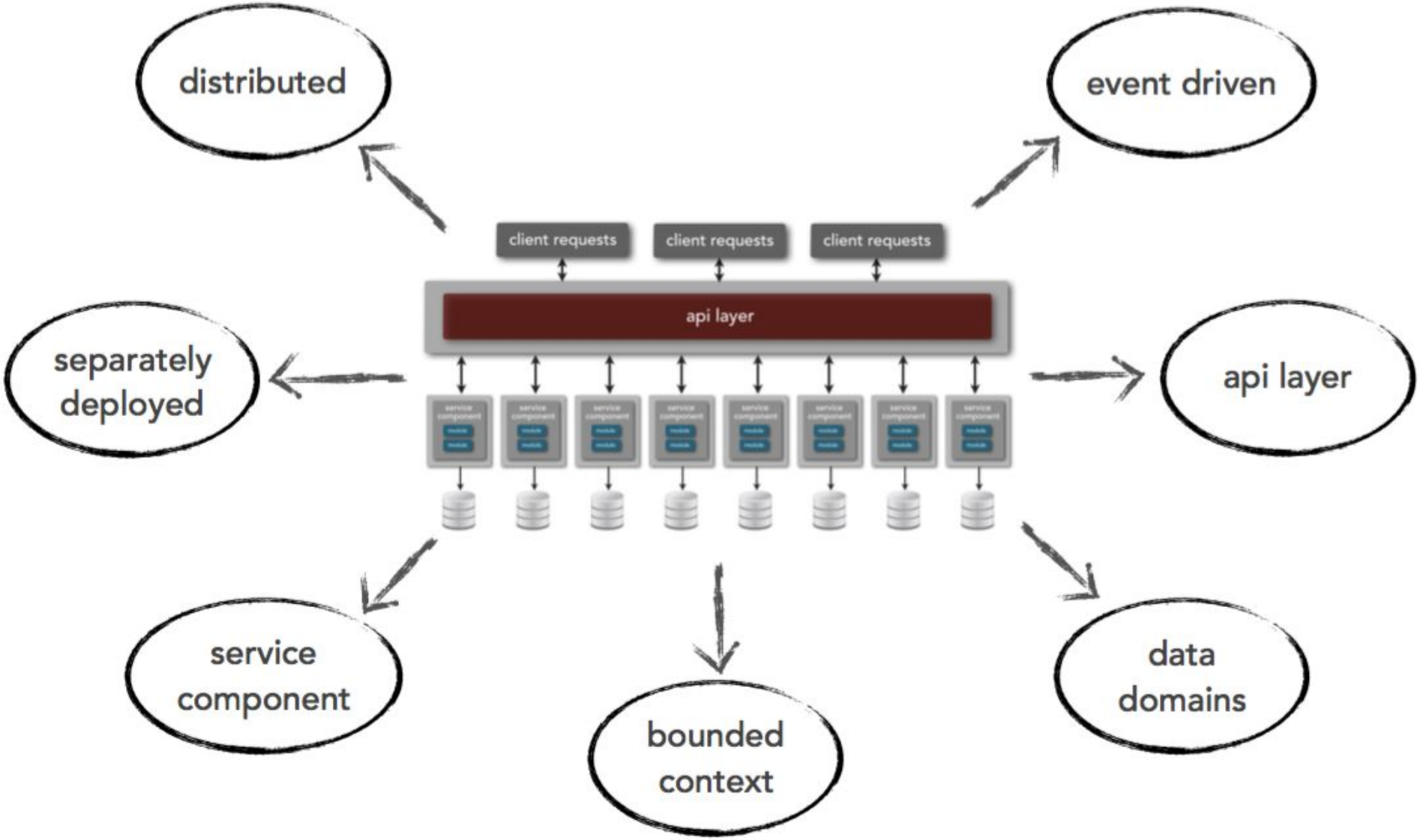
Space-based Architecture

	agility	deployment	testability	performance	scalability	simplicity	cost
	👎	👎	👍	👎	👎	👍	$
	👍	👍	👍	👎	👎	👎	$$
	👍	👍	👎	👍	👍	👎	$$$
	👍	👎	👍	👎	👎	👍	$
	👍	👍	👎	👍	👍	👎	$$$$

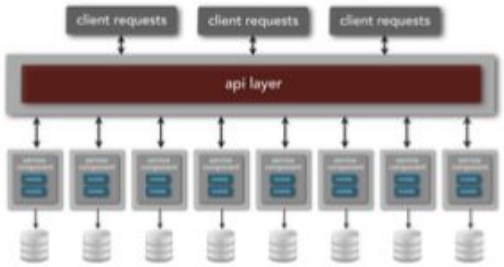
Microservice Architecture



Microservice Architecture



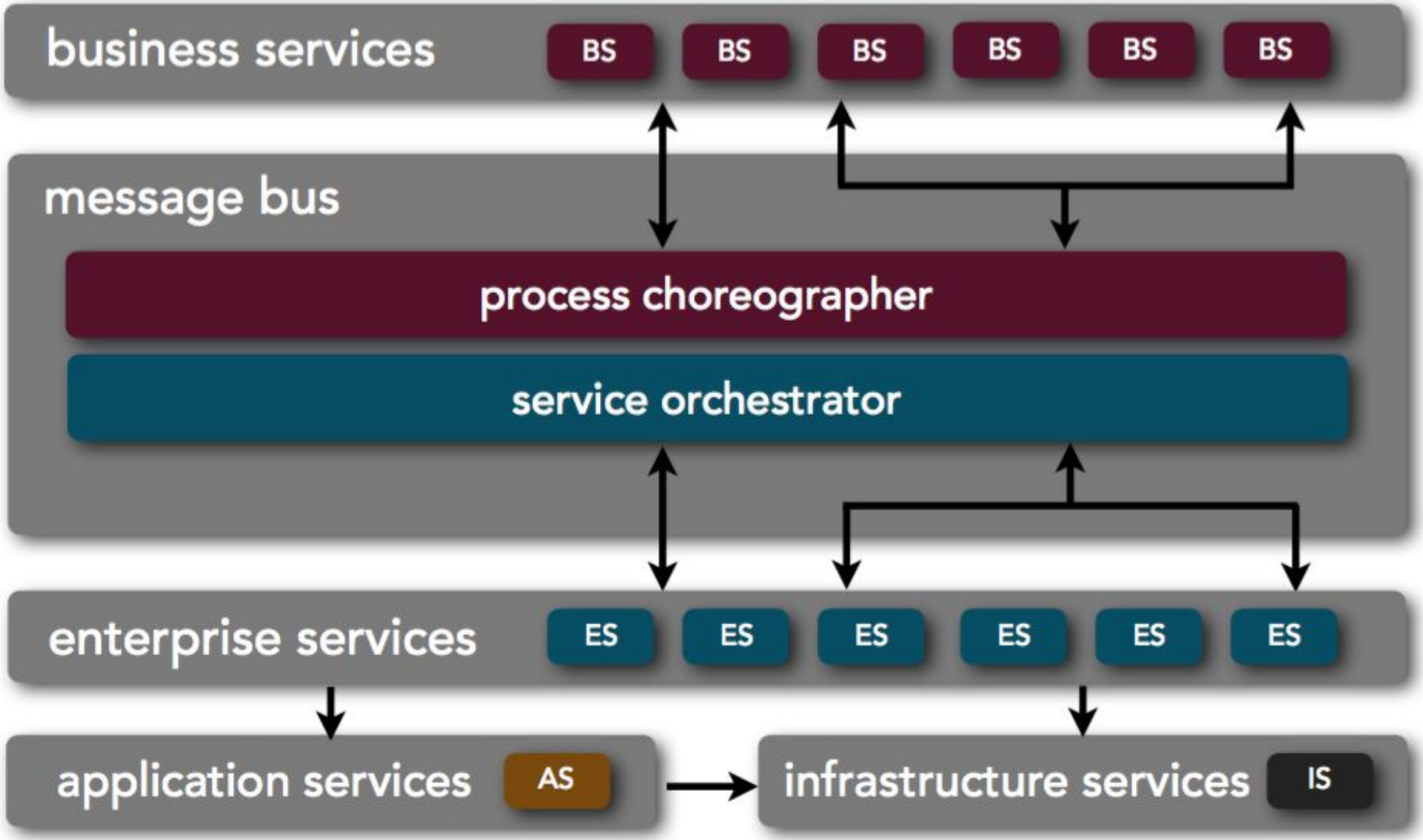
Microservice Architecture

SERVICE REGISTRY AND DISCOVERY	SECURITY AND COMPLIANCE	LOAD BALANCING
DEPLOYMENT	INTER-SERVICE COMMUNICATION	NETWORK
MONITORING		INFRASTRUCTURE AUTOMATION
CONTINUOUS INTEGRATION		PLATFORM MANAGEMENT
CONTAINER REGISTRY	API MANAGEMENT	SERVICE OPTIMIZATION
CLOUD MANAGEMENT	OPERATING SYSTEM	DATABASE MANAGEMENT

Microservice Architecture

	agility	deployment	testability	performance	scalability	simplicity	cost
	👎	👎	👍	👎	👎	👍	$
	👍	👍	👍	👎	👎	👎	$$
	👍	👍	👎	👍	👍	👎	$$$
	👍	👎	👍	👎	👎	👍	$
	👍	👍	👎	👍	👍	👎	$$$$
	👍	👍	👍	👎	👍	👎	$$$

Service-oriented Architecture

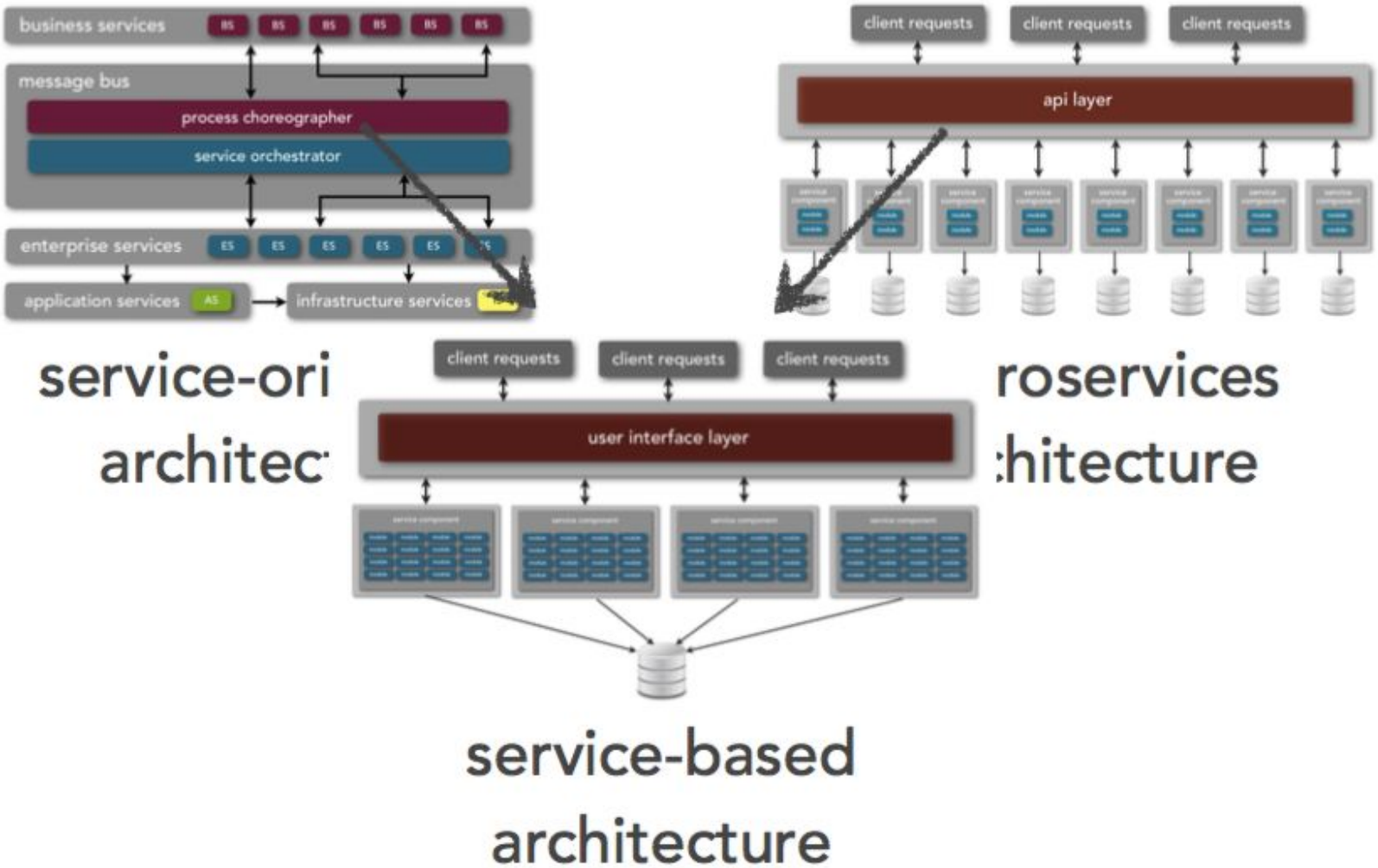


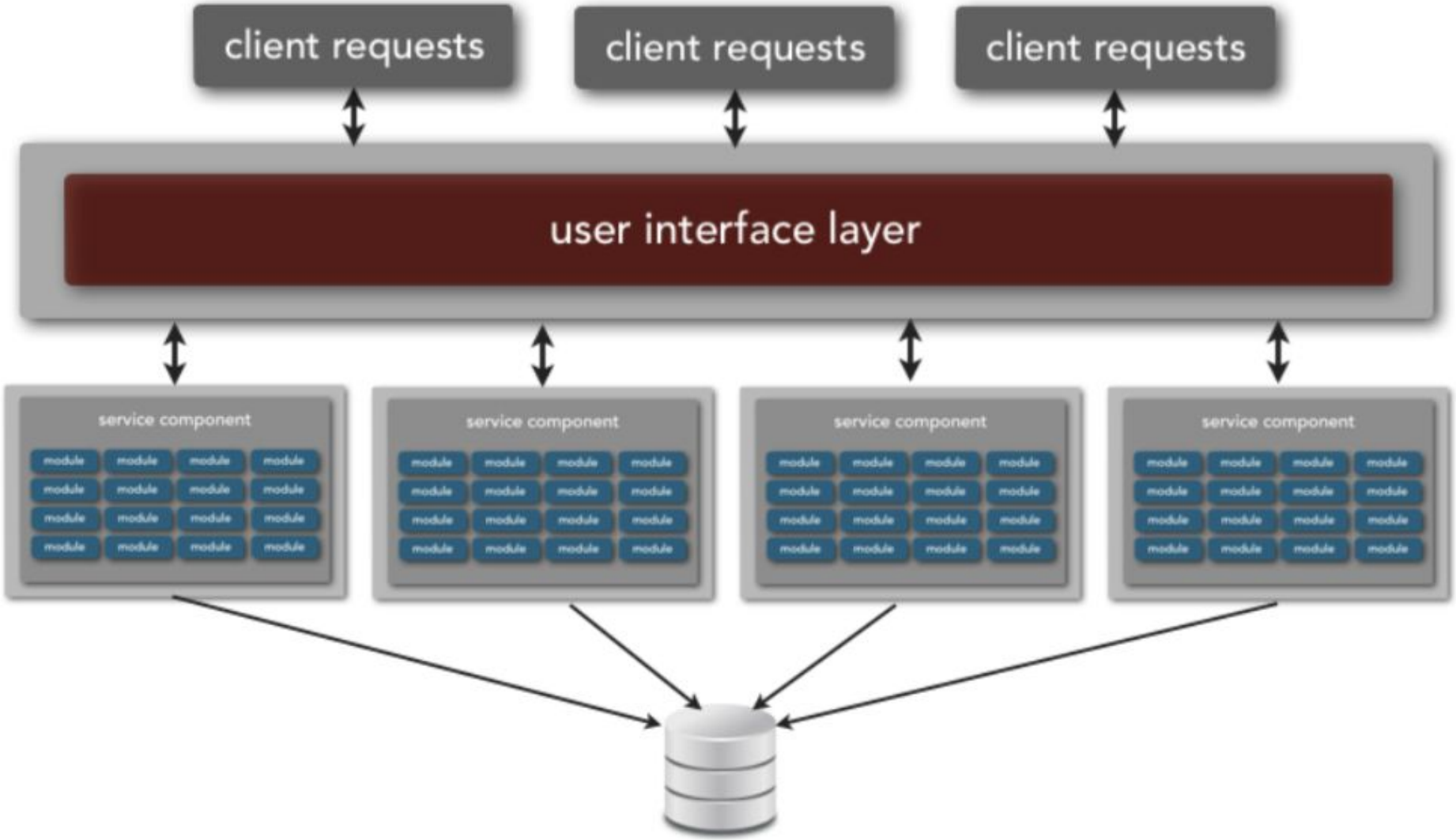
Service-oriented Architecture

	agility	deployment	testability	performance	scalability	simplicity	cost
	👎	👎	👍	👎	👎	👍	$
	👍	👍	👍	👎	👎	👎	$$
	👍	👍	👎	👍	👍	👎	$$$
	👍	👎	👍	👎	👎	👍	$
	👍	👍	👎	👍	👍	👎	$$$$
	👍	👍	👍	👎	👍	👎	$$$
	👎	👎	👎	👎	👍	👎	$$$$

Service-based Architecture

- A middle ground approach




client requests
client requests
client requests
user interface layer
service component
service component
service component
service component

Service-based Architecture

	agility	deployment	testability	performance	scalability	simplicity	cost
	✗	✗	✓	✗	✗	✓	$
	✓	✓	✓	✗	✗	✗	$$
	✓	✓	✗	✓	✓	✗	$$$
	✓	✗	✓	✗	✗	✓	$
	✓	✓	✗	✓	✓	✗	$$$$
	✓	✓	✓	✗	✓	✗	$$$
	✗	✗	✗	✗	✓	✗	$$$$
	✓	✓	✓	✗	✓	✗	$$

Video

https://www.youtube.com/watch?v=BrT3AO8bVQY&t=8s